



UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF M.Sci. WITH HONOURS

FOURTH-YEAR FINAL EXAMINATION

03 00563

LM GENERAL THEORY OF RELATIVITY

SUMMER EXAMINATIONS 2025

Time allowed: 2 hours

Answer four questions from Section 1 and two questions from Section 2.

Section 1 consists of four questions and carries 40% of the marks for the examination.

Answer ***all four*** questions from this Section.

Section 2 consists of three questions and carries 60% of the marks.

Answer ***two*** questions from this Section. If you answer more than two questions, credit will only be given for the *best* two answers.

The approximate allocation of marks to each part of a question is shown in brackets [].

All symbols have their usual meanings.

A list of formulae that may be required can be found at the end of this question paper.

No calculator is permitted in this examination.

SECTION 1

Answer **all four** questions from this Section.

1. State Einstein's equivalence principle and explain how it differs from the weak equivalence principle. [10]

2. A satellite is in a circular orbit of radius $2R$ around the Earth, where R is the Earth radius. At some moment, an atomic clock on the satellite is synchronised with an identical clock on earth. After one revolution it is at some time **ahead** of the clock on Earth. Find the radius of another satellite orbit where the clock would be the same time **behind** the clock on Earth after one revolution. [10]

[Hint: Use the conservation law, $\mathcal{L} = c^2$, for the geometric Lagrangian corresponding to the Schwarzschild metric (see Formula Sheet), Kepler's law for a circular orbit, $\omega^2 r^3 = GM$, and neglect Earth's rotation.]

3. The metric in a two-dimensional space has the form

$$ds^2 = dz^2 + \frac{1}{z^2} d\phi^2, \quad 0 \leq \phi < 2\pi.$$

Write down the corresponding geometric Lagrangian. Use the conservation laws to derive **first-order** differential equations for ϕ and for x and solve them finding expressions for $z(s)$ and $\phi(s)$ that define geodesic lines in this space. [10]

4. A certain two-dimensional space is characterised by the metric

$$ds^2 = d\theta^2 + \sin^2\theta d\phi^2,$$

where $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$.

- (a) Use any method to determine all the nonvanishing Christoffel symbols $\Gamma_{\mu\nu}^\lambda$. [4]

- (b) Find all components of the covariant derivative $v^\mu{}_{;\nu}$ of a constant vector $v = (\frac{\pi}{4}, \pi)$. [6]

SECTION 2

Answer any **two** questions from this Section. If you answer more than **two** questions, credit will only be given for the **best** two answers.

5. A uniformly accelerated motion along the x -axis is convenient to describe in a frame Σ defined by a timelike coordinate u and a spacelike coordinate w which are related to coordinates ct and x in an inertial laboratory frame Σ_0 by

$$w \cosh u = x, \quad w \sinh u = ct.$$

- (a) Show that the Minkowski metric in the new coordinates takes the form

$$ds^2 = w^2 du^2 - dw^2, \quad [4]$$

where the redundant coordinates y and z are suppressed.

- (b) Calculate all the nonzero Christoffel symbols $\Gamma_{\mu\nu}^\lambda$ in Σ (where $\mu, \nu, \lambda = 0, 1$ with the identification $x^0 \equiv u$ and $x^1 \equiv w$). Thus find the scalar curvature R and comment on your result. [9]

A clock $\mathcal{A}$ is at rest in Σ at a point with spatial coordinate w_0 , while another clock, $\mathcal{B}$, is at rest in the laboratory frame Σ_0 at a point $x_0 > 0$ coinciding with w_0 at the moment $t = 0$.

- (c) Show that $x(t)$, the worldline of clock $\mathcal{A}$ in frame Σ_0 , is a hyperbola, identify its asymptotes and sketch it on the $x - ct$ plane together with the worldline of clock $\mathcal{B}$. [6]
- (d) Show that in the non-relativistic limit, $ct \ll w_0$, the worldline of clock $\mathcal{A}$ is reduced to $x(t) = w_0 + \frac{1}{2}at^2$, and express the acceleration a via w_0 . [3]
- (e) Show that the proper time τ at clock $\mathcal{A}$ is given by $c\tau = w_0 u$ and find its relation to t , the time at clock $\mathcal{B}$. [2]
- (f) Explain why the asymptotes found in (c) are “event horizons” in frame Σ and identify the time t_0 at clock $\mathcal{B}$ after which this clock cannot be observed in this frame. [6]

6. A satellite is moving in a circular orbit in the equatorial plane around a neutron star of radius $R=3r_s$. It has angular momentum $J=\sqrt{\frac{27}{8}}cr_s$ (in the units where its mass $m=1$) and energy $\mathcal{E}$, which you will need to determine.

- (a) Using the Schwarzschild metric (see Formula Sheet), write down the corresponding geometric Lagrangian $\mathcal{L}$, identify the cyclic variables and show that the corresponding conservation laws can be written as

$$\mathcal{E} = c^2 \sigma \dot{t}, \quad J = r^2 \dot{\phi}, \quad \mathcal{L} = c^2. \quad [6]$$

- (b) Use the conservation laws to show that the square of the proper radial velocity of the particle is given by

$$\dot{r}^2 = \frac{\mathcal{E}^2}{c^2} - \sigma(r) \left(c^2 + \frac{J^2}{r^2} \right). \quad [4]$$

- (c) Use the result of part (b) to determine

i. the radius of the satellite orbit [8]

ii. its full energy $\mathcal{E}$. [4]

[Hint: Explain why in a circular orbit both $\dot{r} = 0$ and $\ddot{r} = 0$. and use these relations.]

- (d) A laser pulse is sent from the satellite to a stationary remote observer once each orbit, when they are on the same radial line. Calculate the time between the two consecutive signals as measured on the satellite and by the observer. [8]

7. The metric of a 3-dimensional space is given by

$$ds^2 = dr^2 + f^2(r) [d\theta^2 + \sin^2 \theta d\phi^2],$$

where $r \equiv x^1$, $\theta \equiv x^2$, and $\phi \equiv x^3$ are spherical polar coordinates and $f(r)$ is a smooth function.

(a) Write down the metric tensor, $g_{\mu\nu}$, and its inverse, $g^{\mu\nu}$. [3]

(b) Explain why Γ_{22}^1 , Γ_{33}^1 , $\Gamma_{12}^2 = \Gamma_{21}^2$, and $\Gamma_{13}^3 = \Gamma_{31}^3$ are the only nonzero Christoffel symbols and calculate them using your preferred method. [9]

(c) Show that scalar curvature R can be written via components of the Riemann tensor as

$$R = 2(R^{12}_{12} + R^{13}_{13} + R^{23}_{23}).$$
 [3]

(d) Show that

$$R^{23}_{23} = \frac{1 - (f')^2}{f^2}, \text{ where } f' \equiv \frac{df(r)}{dr},$$

and calculate the other two components of the Riemann tensor required for the scalar curvature R . [9]

(e) Express R in terms of $f(r)$ and its derivatives. Show that if $f(r) = \sin r$, or r , or $\sinh r$, then R is a constant and calculate its value in each case. Hence comment on the importance of this metric for cosmology. [6]

Formula Sheet

Unless otherwise stated, all indices take values of $\mu = (0, 1, 2, 3)$ in the GR context, or $\mu = (1, \dots, d)$ in a d -dimensional space, and Einstein's summation convention is implied

Metric and Lagrangian

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu, \quad \mathcal{L} \equiv g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu, \quad \dot{x}^\mu \equiv u^\mu \equiv \frac{dx^\mu}{ds}, \quad g_{\mu\nu} \text{ is a metric tensor}$$

$$\text{In GR context, } s \mapsto \tau, \text{ i.e. } \dot{x}^\mu \equiv \frac{dx^\mu}{d\tau} \text{ where } \tau \text{ is the proper time, } ds^2 = c^2 d\tau^2$$

The Schwarzschild metric

$$ds^2 = c^2 \sigma(r) dt^2 - \sigma^{-1}(r) dr^2 - r^2 [d\theta^2 + \sin^2\theta d\phi^2], \quad \sigma(r) \equiv \left(1 - \frac{r_s}{r}\right), \quad r_s \equiv \frac{2GM}{c^2}$$

Christoffel symbols, $\Gamma_{\alpha\beta}^\mu$

$$\Gamma_{\alpha\beta}^\mu \equiv g^{\mu\nu} \Gamma_{\nu\alpha\beta}, \quad \Gamma_{\nu\alpha\beta} = \frac{1}{2} (g_{\nu\alpha,\beta} + g_{\nu\beta,\alpha} - g_{\alpha\beta,\nu})$$

Covariant derivatives

$$\nabla_\beta \xi^\mu \equiv \xi^\mu_{;\beta} = \xi^\mu_{,\beta} + \Gamma_{\alpha\beta}^\mu \xi^\alpha, \quad \nabla_\beta \xi_\mu \equiv \xi_{\mu;\beta} = \xi_{\mu,\beta} - \Gamma_{\mu\beta}^\alpha \xi_\alpha, \quad \xi^\mu_{,\beta} \equiv \partial_\beta \xi^\mu \equiv \frac{\partial \xi^\mu}{\partial x^\beta}$$

Covariant dot operation

$$\dot{\xi} \equiv \frac{d}{ds} (\xi^\alpha \mathbf{e}_\alpha) = (\nabla_u \xi^\alpha) \mathbf{e}_\alpha \equiv (\xi^\alpha_{,\mu} + \Gamma_{\nu\mu}^\alpha u^\nu) u^\mu \mathbf{e}_\alpha \text{ where } \nabla_u \equiv u^\mu \nabla_\mu$$

Four-acceleration f^α and geodesic equations in the Euler–Lagrange and canonical form

$$f^\alpha = \frac{g^{\alpha\beta}}{2} \left[\frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\beta} \right) - \frac{\partial \mathcal{L}}{\partial x^\beta} \right] = \ddot{x}^\alpha + \Gamma_{\mu\nu}^\alpha \dot{x}^\mu \dot{x}^\nu$$

$$f^\alpha = 0 \Rightarrow \dot{u}^\alpha + \Gamma_{\mu\nu}^\alpha u^\mu u^\nu = 0 \quad \Leftrightarrow \quad \nabla_u u^\alpha = \nabla_u^2 x^\alpha = 0.$$

The Riemann, $R_{\nu\alpha\beta}^\mu$

$$R_{\nu\alpha\beta}^\mu \xi^\nu = \xi^\mu_{;\alpha\beta} - \xi^\mu_{;\beta\alpha}, \quad R_{\nu\alpha\beta}^\mu = -\Gamma_{\nu\alpha,\beta}^\mu + \Gamma_{\nu\beta,\alpha}^\mu - \Gamma_{\nu\alpha}^\sigma \Gamma_{\beta\sigma}^\mu + \Gamma_{\nu\beta}^\sigma \Gamma_{\alpha\sigma}^\mu.$$

The Riemann in locally (pseudo)Cartesian frame (LCF)

$$g_{\mu\nu} R_{\alpha\beta\gamma}^\mu = R_{\nu\alpha\beta\gamma} = g_{\nu\gamma,\alpha\beta} - g_{\nu\beta,\alpha\gamma} + g_{\alpha\beta,\nu\gamma} - g_{\alpha\gamma,\nu\beta}.$$

Symmetries of the Riemann tensor

$$R^\alpha{}_{\beta\mu\nu} + R^\alpha{}_{\mu\nu\beta} + R^\alpha{}_{\nu\beta\mu} = 0, \quad R^\alpha{}_{\beta\mu\nu} = -R^\alpha{}_{\nu\mu\beta}, \quad R_{\alpha\beta\mu\nu} = R_{\mu\nu\alpha\beta}.$$

Geodesic deviation

$$f^\alpha = \nabla_u^2 x^\alpha(\tau; \eta_0 + d\eta) = R^\alpha{}_{\beta\mu\nu} u^\beta u^\mu \xi^\nu, \quad u^\mu = \frac{dx^\mu(\tau; \eta_0)}{d\tau}, \quad \xi^\nu = \frac{dx^\nu(\tau; \eta)}{d\eta} d\eta.$$

The mixed Ricci, $R^\mu{}_\nu$, and Einstein, $G^\mu{}_\nu$, tensors and scalar curvature, R

$$R^\mu{}_\nu \equiv R^{\alpha\mu}{}_{\alpha\nu}, \quad G^\mu{}_\nu \equiv R^\mu{}_\nu - \frac{1}{2} \delta^\mu{}_\nu R, \quad R = \delta^\nu{}_\mu R^\mu{}_\nu.$$

Einstein's equations

$$G^\mu{}_\nu = \kappa T^\mu{}_\nu; \quad \kappa \equiv \frac{8\pi G}{c^4}, \quad T^\mu{}_\nu \text{ is a matter tensor.}$$

In the cosmological context, $\Lambda \delta^\mu{}_\nu$ should be added to the r.h.s., where $\Lambda > 0$ is the cosmological constant.

The matter tensor in the perfect fluid model of the Universe

$$T^\mu{}_\nu = (\rho_M + \rho_R) u^\mu u_\nu - P \delta^\mu{}_\nu.$$

Here $\rho_{M(R)}$ is the matter (radiation) density, with $\rho_R \equiv 3P/c^2$ where P is the radiation pressure.

The Friedmann–Robertson–Walker metric

$$ds^2 = c^2 dt^2 - \mathcal{R}^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2\theta d\phi^2) \right].$$

Here $k = 1, 0$, or -1 is the curvature parameter, and $\mathcal{R}(t)$ is the history function.

The Hubble ‘constant’ H and ‘deceleration’ parameter q

$$H \equiv \frac{\dot{\mathcal{R}}}{\mathcal{R}}, \quad q \equiv -\frac{\ddot{\mathcal{R}}\mathcal{R}}{\dot{\mathcal{R}}^2}, \quad \text{where} \quad \dot{\mathcal{R}} \equiv \frac{d\mathcal{R}}{dt}, \quad \ddot{\mathcal{R}} \equiv \frac{d^2\mathcal{R}}{dt^2}.$$

Cosmological density parameter

$$\Omega = \Omega_M + \Omega_R + \Omega_V = \frac{\rho_M + \rho_R + \rho_V}{\rho_{\text{crit}}}, \quad \text{where} \quad \rho_V \equiv \frac{\Lambda c^2}{8\pi G} \quad \text{and} \quad \rho_{\text{crit}} \equiv \frac{3H^2}{8\pi G}.$$

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.